Solving Absolute Value Inequalities

The "In or Out" Method

The Big Rule

Absolute value bars ask: how far is the inside expression from 0? Instead of memorizing which symbol means *and* and which means *or*, **test the middle point** (where the inside expression equals 0) to see whether it belongs to the solution:

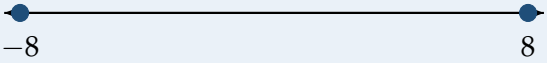
Test result at the middle	Region	Compound inequality
True	**in** (between the boundaries)	**and**
False	**out** (outside the boundaries)	**or**

This works because $|\text{expr}| \le k$ describes a single connected interval around 0 (an *and*), while $|\text{expr}| \ge k$ describes everything *except* that interval (an *or*).

Example 1 Solve $|3x - 4| \le 8$ (an "And" case)

Step 1 Find the boundaries

Since $|-8| = 8$ and $|8| = 8$, the expression $3x - 4$ hits the edge at -8 and 8. Mark both as **closed** dots (the inequality is $\le$, so the ends are included).



Step 2 Check: In or Out?

Test the middle point $3x - 4 = 0$:

$$|0| = 0 \le 8 \quad \textbf{True}.$$

The point *between* the boundaries works, so we keep the **in** region. That makes this an **and** statement (a *between* inequality), not an *or*.

Step 3 Write the compound inequality and solve

Because it is "in," $3x - 4$ stays *between* -8 and 8:

$$-8 \le 3x - 4 \le 8.$$

Solve all three parts at once. Add 4:

$$-8 + 4 \le 3x - 4 + 4 \le 8 + 4 \qquad \Longrightarrow \qquad -4 \le 3x \le 12.$$

Divide by 3:

$$\frac{-4}{3} \le x \le \frac{12}{3} \qquad \Longrightarrow \qquad -\frac{4}{3} \le x \le 4.$$

Solution: $-\frac{4}{3} \le x \le 4$ or $\left[-\frac{4}{3}, 4\right]$



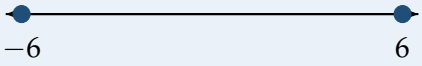
Check: at $x = 0$, $|3(0) - 4| = 4 \le 8$ ✓

Example 2 Solve $|2x + 4| \ge 6$ (an "Or" case)

Same method as Example 1 – but this time the test point will fail, which flips it to an *or* statement.

Step 1 Find the boundaries

Since $|-6| = 6$ and $|6| = 6$, the expression $2x + 4$ hits the edge at -6 and 6. Mark both as **closed** dots (the inequality is $\ge$, so the ends are included).



Step 2 Check: In or Out?

Test the middle point $2x + 4 = 0$:

$$|0| = 0 \ge 6 \quad \textbf{False}.$$

The point *between* the boundaries does **not** work, so the solution is the **out** region. That makes this an **or** statement (an *outside* inequality), not an *and*.

Step 3 Write the compound inequality and solve

Because it is "out," $2x + 4$ stays *outside* -6 and 6 — less than or equal to the left boundary, **or** greater than or equal to the right boundary:

$$2x + 4 \le -6 \quad \textbf{or} \quad 2x + 4 \ge 6.$$

Solve each piece on its own. Subtract 4 from both sides:

$$2x \le -10 \quad \textbf{or} \quad 2x \ge 2.$$

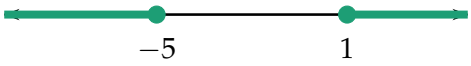
Divide both sides by 2:

$$x \le -5 \quad \textbf{or} \quad x \ge 1.$$

Common Mistake to Avoid

The original inequality is $\geq$, not a strict $>$ – so the boundary points themselves are part of the solution, and the compound inequality must use $\leq$ and $\geq$, not $<$ and $>$. Writing "$2x+4 < -6$ or $2x+4 > 6$" would quietly kick $x = -5$ and $x = 1$ back out of the answer. Always check: plug the boundary value into the *original* inequality — at $x = -5$, $|2(-5)+4| = |-6| = 6 \geq 6$ ✓, so it belongs, and the dots on the number line stay **closed**.

Solution: $x \leq -5$ or $x \geq 1$ or $(-\infty, -5] \cup [1, \infty)$



Check: at $x = -10$ (outside), $|2(-10)+4| = 16 \geq 6$ ✓ at $x = 0$ (between), $|2(0)+4| = 4 \geq 6$? ✗ (correctly excluded)

Key Idea — And vs. Or at a Glance

Form	Test at the middle	Solution shape
$\lvert\text{expr}\rvert \leq k$	always True (In)	**and**: one connected interval
$\lvert\text{expr}\rvert \geq k$	always False (Out)	**or**: two separate rays

(Assuming $k > 0$.) The middle-point test is a shortcut for the graph of $y = |\text{expr}|$: it dips *below* the line $y = k$ only in the middle (giving an *and*), so anywhere it stays at or above $y = k$ – like $|2x+4|$ compared to $y = 6$ in Example 2 – gives an *or*.